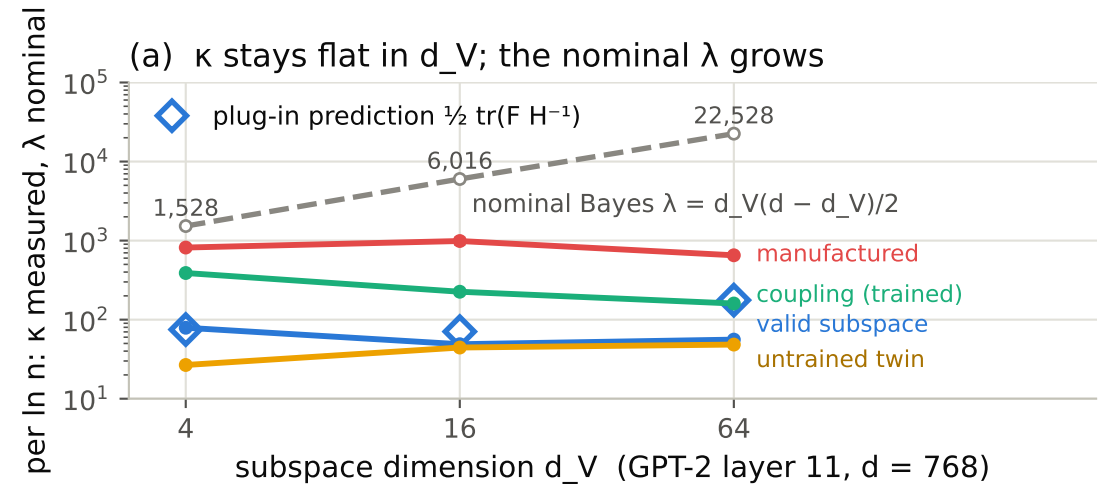
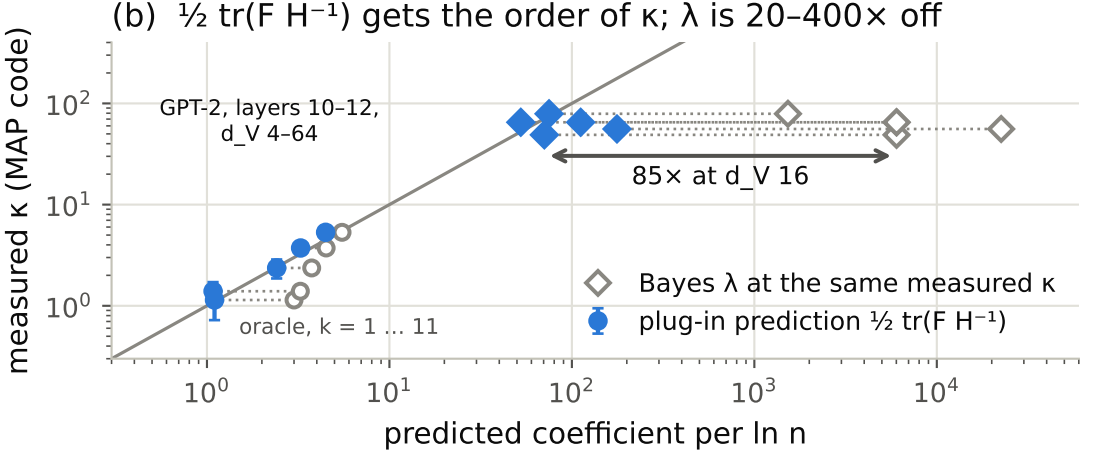
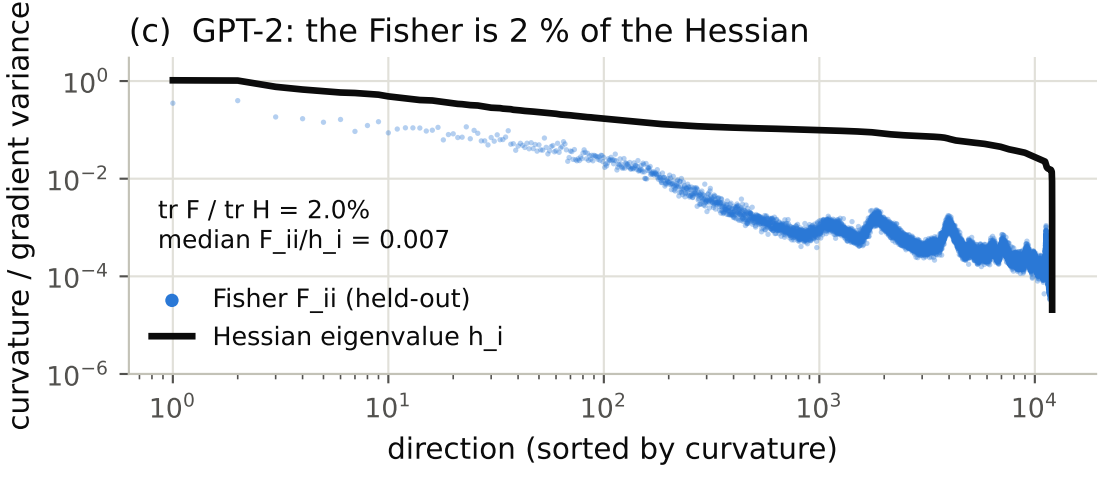
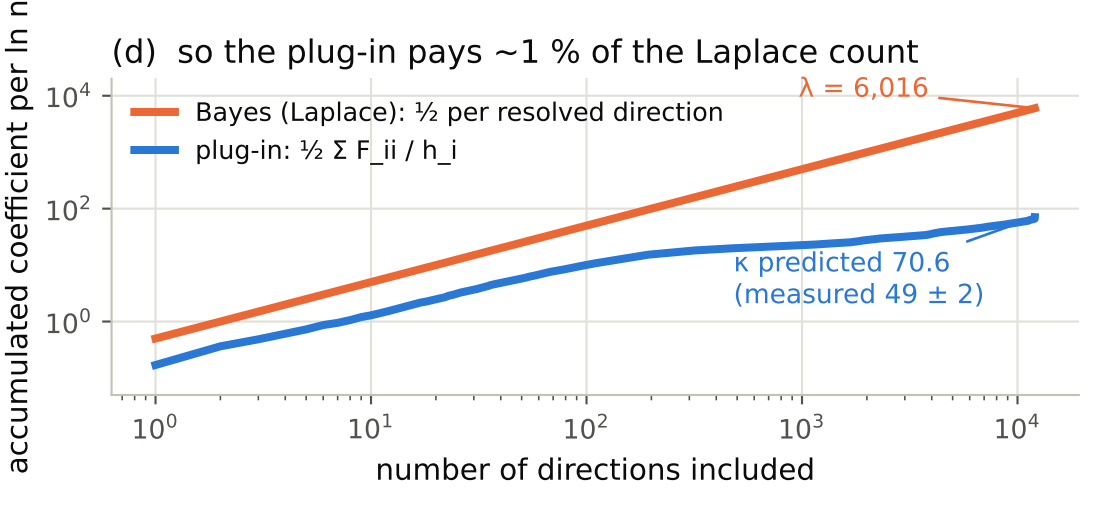

(a) κ stays flat in d_V; the nominal λ grows
per ln n: κ measured, λ nominal
plug-in prediction ½ tr(F H⁻¹)
22,528
6,016
1,528
nominal Bayes λ = d_V(d − d_V)/2
manufactured
coupling (trained)
valid subspace
untrained twin
subspace dimension d_V (GPT-2 layer 11, d = 768)


(b) ½ tr(F H⁻¹) gets the order of κ; λ is 20–400× off
measured κ (MAP code)
GPT-2, layers 10–12,
d_V 4–64
85× at d_V 16
oracle, k = 1 … 11
Bayes λ at the same measured κ
plug-in prediction ½ tr(F H⁻¹)
predicted coefficient per ln n


(c) GPT-2: the Fisher is 2 % of the Hessian
curvature / gradient variance
tr F / tr H = 2.0%
median F_ii/h_i = 0.007
Fisher F_ii (held-out)
Hessian eigenvalue h_i
direction (sorted by curvature)


(d) so the plug-in pays ~1 % of the Laplace count
accumulated coefficient per ln n
λ = 6,016
Bayes (Laplace): ½ per resolved direction
plug-in: ½ Σ F_ii / h_i
κ predicted 70.6
(measured 49 ± 2)
number of directions included